# Haocheng He

haocheng.he@foxmail.com | Hong Kong SAR, China | linkedin.com/in/haocheng-he-92657028b

## PROFESSIONAL SUMMARY

Industrial Engineering graduate with hands-on experience in process optimization, simulation modeling, and AI deployment across four internships. Background combines factory-floor time studies, airport operations research, and hardware/software RD. Holds a law minor providing regulatory awareness. Seeking roles in Operations Analytics and Technology Risk.

## EDUCATION

**The Hong Kong Polytechnic University (PolyU)** — Hong Kong SAR, China
*Master of Engineering in Aviation Engineering and Operation Management* — Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)

**Guangdong University of Technology (GDUT)** — Guangzhou, China
*Bachelor of Engineering in Industrial Engineering* — Sep 2021 – Jun 2025
GPA: 3.67 / 4.0 (Top 20%) – Excellent Graduation Thesis + Innovation Award

*Minor in Law* — Sep 2022 – Jun 2025
GPA: 3.61 / 4.0

## EXPERIENCE

**AI Operations & R&D Intern** — May 2025 – Apr 2026
*Shenzhen Wuxian Hongyin Technology Co., Ltd.* — *Shenzhen, China*

- Deployed AI inference workflows on Huawei Ascend 310B chips for internal team use; sped up response time by approximately 30%.
- Designed motor driver circuits (LCEDA) and wrote STM32 firmware for a smart piano; tested with oscilloscopes and logic analyzers.
- Built a finger-position tracking system for the piano using a camera and the Ascend chip; documented the SDK for the team.
- Managed servers (Ubuntu), network infrastructure (Huawei switches and routers), procurement, and interviewed candidates for new roles.

**Research Assistant – Airport Operations Simulation** — Sep 2023 – Jun 2025
*Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport)* — *Guangzhou, China*

- Modeled ground vehicle operations at Guangzhou Baiyun Airport using AnyLogic, Java, and MySQL – roads, service lanes, and dispatch rules.
- Implemented routing algorithms (Dijkstra, A*, BFS, genetic) for 50+ vehicles across 3 zones to find collision-free paths.
- Reduced simulated congestion by 35% during peak hours (200+ movements/hour) with zero collisions.
- Thesis won Excellent Thesis + Innovation Award; the airport team now uses the model for planning.

**Industrial Engineering Intern** — Apr 2024 – May 2024
*Midea Group – Kitchen & Water Heater Division* — *Foshan, China*

- Timed every step on a dishwasher sheet metal line by hand; found 5 steps that exceeded cycle time.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

**Manufacturing Operations Intern** — Jul 2023 – Aug 2023
*Guangxi LiuGong Machinery Co., Ltd.* — *Liuzhou, China*

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Discussed findings with line engineers.

## SKILLS

**Programming:** Python, Java, SQL, STM32 (C)
**Tools & Platforms:** AnyLogic, Plant Simulation, OR-Tools, MySQL, LCEDA, oscilloscope, logic analyzer
**Languages:** English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

## PUBLICATIONS & INTELLECTUAL PROPERTY

- 3 Invention Patents: PI Coating Machine Motion Control via MCMC Sampling (CN122085794A); Coating Control Method and System – Deep Reinforcement Learning (CN121325552A); Home Solar Energy Storage System – Adaptive Energy Regulation (CN117767437A).
- 1 Utility Model: Integrated Wire Harness Production Equipment (CN221352455U).
- 2 Software Copyrights: Enterprise Financial Data Analysis Software V1.0 (2024SR0146827); Urban Flood Intelligent Mobile Emergency Command Dispatch System V1.0 (2024SR0549247).
- Excellent Bachelor Thesis and Innovation Award – Guangdong University of Technology, 2025.